KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2021
Department of Computer Science and Engineering
CSE 4127
Image Processing & Computer Vision

TIME: 3 hours | FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

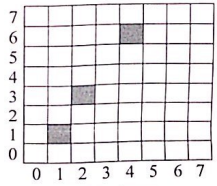
1. a) Derive the equation for Laplacian of a 2D Gaussian function $G = A \cdot e^{-\frac{x^2+y^2}{2\sigma^2}}$. Sample the LoG function to get a convolution kernel with $A = \sigma^2 = 1$. Scale the kernel values to the nearest integer. Use appropriate kernel size to support the function. (17)

b) Determine the 4-point 1D DFT matrix. Use it to find the 2D DFT of the following image. (13)

$$\begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & -1 \\ -1 & 2 & 1 & 0 \\ 2 & -1 & 1 & 1 \end{bmatrix}$$

c) Explain an implementation idea for hysteresis thresholding used in Canny edge detection method. (05)

2. a) Use Hough line transformation for the following edge image to detect the line. Use an appropriate accumulator array with $d\theta = 30°$, $d\rho = 1$. θ spans from 0° to 180°, ρ spans from -4 to 8. Show vote count in the array and determine the line equation. (18)



Edge Image

b) Use weighted median filter to restore the following 4-bit image. Show results only for the middle row, use the given weight matrix. (10)

4	3	1	2	6
2	7	3	5	0
9	15	6	0	3
12	4	10	8	7
11	6	2	7	6

Image

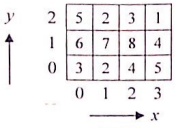
1	2	1
2	3	2
1	2	1

Weight Matrix

c) If $x_0(t) = at/T$ and $y_0(t) = bt/T$ denotes the time varying components of motion in x and y directions, then show that uniform linear motion between image and the sensor during image acquisition can be modeled by the following degradation function. Here, symbols have their usual meanings. (07)

$$H(u,v) = \frac{T}{\pi(ua+vb)} \sin[\pi(ua+vb)] e^{-j\pi(ua+vb)}$$

3. a) What is the importance of image warping in computer vision? Warp the following image with the given transformation matrix, T. Use backward mapping and bilinear interpolation during the process. Mention the size of the warped image and determine the intensity only at pixel (2, 3). (16)



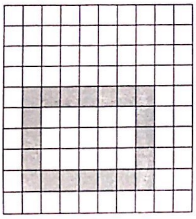
Image

$$T = \begin{bmatrix} 0.43 & -0.25 \\ 1 & 1.73 \end{bmatrix}$$

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b) State the maximum entropy based image thresholding algorithm with necessary mathematical derivation. (16)

c) Draw signature for the following binary image. (03)



Binary Image

4. a) State the pros and cons of histogram based texture descriptors with proper example. (08)

b) For the following image, determine the Gray Level Co-occurrence Matrix (GLCM) with displacement $d = [1, 1]$. And thereby, measure the uniformity and homogeneity of the matrix. (17)

y \ x	0	1	2	4	1
1	4	0	2	3	
1	1	3	2	0	
3	2	0	3	1	
4	3	2	4	2	

c) Derive the equations used by Lucas-Kanade to estimate motion field from a video sequence. (10)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Illustrate the 'Golomb Coding' technique to encode non-negative integer in data compression. Calculate the Golomb Codes for the following representation: (20)

i) $G_{11}(37)$

ii) $G_{12}(-15)$

iii) $G^2_{exp}(13)$

b) Explain false contouring effect in digital image using the concept of sampling and quantization. (08)

c) What measure can be taken to improve the brightness discrimination provided that a weber ratio is given? (07)

6. a) Explain the application of pseudo-color processing for highlighting explosives contained in a luggage using proper transformation functions. (12)

b) Write down the conditions for pre-color-component processing and vector-based processing to be equivalent with proper example. (08)

c) Derive the CMY intensity mapping function $s'_i = kr'_i + (1 - k)$ using its RGB counterpart where $0 < k < 1$. (10)

d) Prove, erosion and dilation are duals of each other with respect to set complementation and reflection. (05)

7. a) Write down the technique for filtering the hue component in HSI color space provided that the saturation is low and hue value is quite unstable. (12)

b) 'It is possible to save a digital image with roughly 50% less storage than the original image using bit plane slicing.' – Justify the statement. (08)

c) Illustrate the objective fidelity criteria for quantifying the nature of loss in image compression. (10)

d) Mention the drawbacks of 'Convex Hull' algorithm. (05)

8. a) Illustrate the automated hole filling algorithm for binary image based on morphological reconstruction. (12)

b) Suppose, you are asked to enhance dark areas of an image that may contain hidden features while leaving the light area as unchanged as possible. Now, apply 'local enhancement by contrast manipulation' to solve this problem using histogram statistics. (15)

c) If histogram equalization is applied twice to an image, that is, if it is applied to the result of histogram equalization, will the second application change the image or not? Explain why or why not? (08)

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